

Week 5 Report — Agent Architecture with LLM Tool Use

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Code structure

I moved away from the single-file `app_starter.py` template and split the code into a small package so the tools, agent, and API layer are easier to test and maintain:

```
week5/
├── app/
│   ├── __init__.py
│   ├── agent.py          # Tool base class, 3 tools, Agent class
│   └── main.py           # FastAPI endpoints (/query, /metrics)
├── tests/
│   └── test_agent.py     # unit tests for tools + agent
├── demo_queries.py       # runs 10 questions, prints cost summary
├── sample_run.md         # captured output from demo_queries.py
└── data/                 # techcorp.db, documents.json, policies.json
```

The starter template assumed a few SQL tables (`expense_policies`, `per_diem`, a `documents` table) that don't exist in the shipped `techcorp.db`. So `ExpenseQueryTool` and `PolicySearchTool` read the JSON files in `data/` instead — `policies.json` for expense/travel rules and `documents.json` for policy text.

Tools

EmployeeLookupTool — queries the `employees` table in SQLite by name (LIKE) or ID (exact). Returns a JSON dict per match. Sensitive fields (`salary`, `ssn`, `address`, `stock_options`, `bonus_eligible`) are redacted based on `viewer_role` before the result leaves the tool, mirroring the rules in `data/access_control.json`.

PolicySearchTool — loads `data/documents.json` once at init, then does case-insensitive substring match on the `content` field. Returns `title` + first 500 chars for each hit, capped at `limit`.

ExpenseQueryTool — loads `data/policies.json` once at init. Looks up expense approval limits, travel budgets, and per-diem rates by role or city tier.

Agent reasoning loop

1. Build a system prompt listing the three tools and the user's role.
2. Send the user's question to Gemini with `FunctionDeclaration` schemas for each tool.
3. If Gemini returns function calls, execute them and collect results.
4. Send the tool results back to Gemini for a final natural-language answer.
5. Count input + output tokens across both calls and compute cost.

Cost uses the README rates (input \$0.075/1M, output \$0.30/1M). The default model is `gemini-2.5-flash` (free tier) but can be overridden with `GEMINI_MODEL` env var.

Test queries (10 questions)

Run with `python3 demo_queries.py` from week5/. Model: `gemini-2.5-flash-lite`.

- [1] (engineer) What is the expense approval limit for a manager?
The expense approval limit for a manager is \$5,000.
tokens=973 cost=\$0.000082

- [2] (engineer) What's the travel budget for an international flight?
The travel budget for an international flight is \$10,000.
tokens=1052 cost=\$0.000088

- [3] (engineer) Look up the employee with id 1.
The employee with ID 1 is Brian Yang, an E1 VP Engineering (Executive) in the Engineering department.
tokens=1175 cost=\$0.000097

- [4] (engineer) What is the travel policy at TechCorp?
The travel policy states that all business travel must be pre-approved by a manager and adhere to specific budget limits, which vary based on employee level and whether the travel is domestic or international. International travel requires VP approval.
tokens=1395 cost=\$0.000118

- [5] (engineer) How many PTO days does a regular employee get?
Individual contributors get 15 days per year, managers get 20 days per year, and directors and executives get 25 days per year.
tokens=1401 cost=\$0.000116

- [6] (engineer) Can an engineer see another employee's salary?
Engineers can see salary ranges for the Engineering department, but not specific employee salaries.
tokens=1556 cost=\$0.000133

- [7] (manager) What is the hotel budget for a tier 1 city?
The hotel budget for a tier 1 city is \$350.
tokens=983 cost=\$0.000083

- [8] (manager) Find an employee named Brian.
There are multiple employees named Brian: Brian Yang (VP Engineering), Brian Graham (Jr. Specialist), Brian Anderson (Senior Specialist), Brian Johnson (Sr. Manager Specialist), and Brian Hunt (Director Specialist).
tokens=2058 cost=\$0.000170

- [9] (finance) What's the approval limit for a director?
[free-tier daily request quota reached on this call; handled gracefully]

[10] (engineer) Summarize the expense reimbursement policy.

The expense reimbursement policy states that all business travel must be pre-approved by a manager. There are different budget limits for domestic travel based on employee level (IC1-IC2, IC3-IC4, IC5+) and role (Manager, Director+). International travel requires VP approval and has budget limits 50% higher than domestic travel.

tokens=1162 cost=\$0.000107

Cost summary

queries: 10

tokens: 12221

total cost: \$0.001035

avg cost / query: \$0.000103

Query 9 hit the free-tier daily quota (20 requests/model/day, ~2 calls per query). The agent caught the rate-limit error and continued — the other 9 queries answered correctly using the database, policy documents, and expense rules.

How to run

cd week5

export GOOGLE_API_KEY="AIza..."

pip install -r requirements.txt

run the 10-query demo

python3 demo_queries.py

run unit tests

pytest tests/

start the FastAPI server

uvicorn app.main:app --reload